

Example – Two-Mass Oscillator

Optimal Control for Trajectory Tracking

Rigid Bodies

- Body_1
- Body_2

Constraints

generic

- Ground <> Body_1
- Ground <> Body_2

Forces/Torques

forces generic

- $F_1 = c_1 x_1 + d_1 v_1$
- $F_2 = c_2 x_2 + d_2 v_2$
- $F_3 = c_3 (x_2 - x_1)$
- u

Measures

- x_1
- v_1
- x_2
- v_2

Data Object

parameter

- f_u

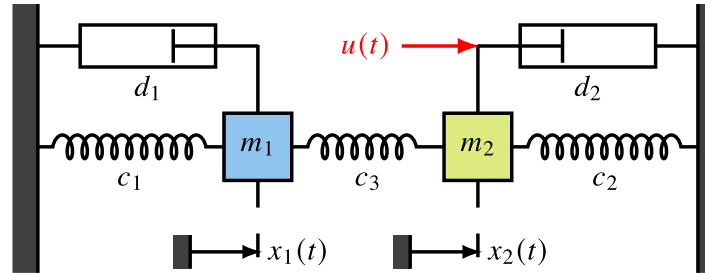
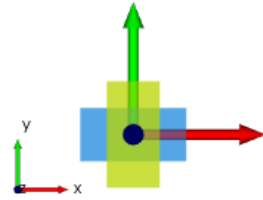
Value: 0

- c_1
- c_2
- c_3
- d_1
- d_2

spline

- $\bar{u}$

.txt

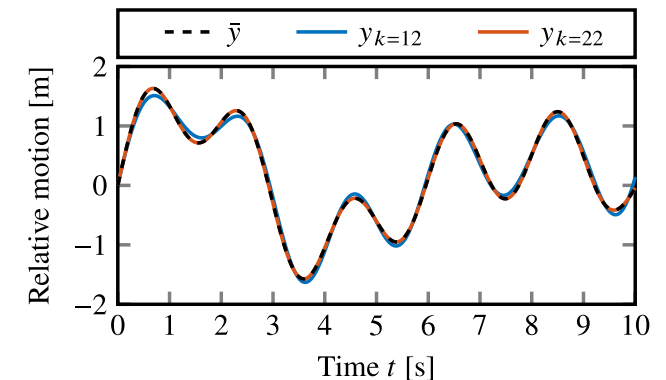
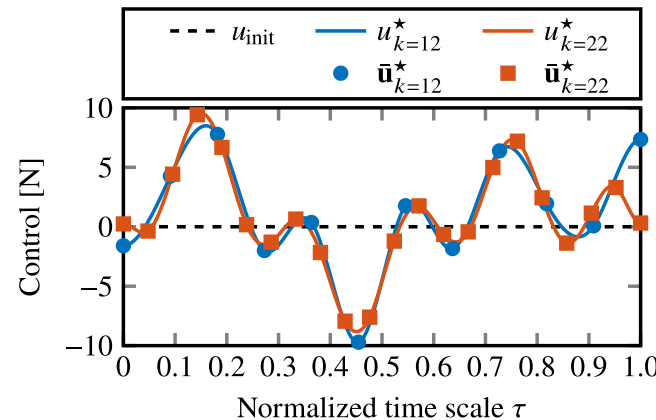


Objective

$$J = \frac{1}{2} \int_{t_0}^{t_f} (y - \bar{y})^2 dt$$

$$\begin{aligned} \bar{y}(t) &= x_2(t) - x_1(t) \\ &= \sin(t) + 0.7 \sin(\pi t) + 0.5 \sin(\sqrt{2} t) \end{aligned}$$

Target Trajectory



Example – Nonlinear Spring Pendulum

Minimum Control Effort during a Rest-to-Rest Maneuver

▢ Rigid Bodies

- Body

▢ Constraints

▢ generic

- Ground <> Body

▢ Forces/Torques

▢ forces generic

- F_x
- F_y
- F_z

$$\mathbf{F}_{\text{Spring}} = -\frac{c}{2l_0^2} (\mathbf{q}^T \mathbf{q} - l_0^2) \mathbf{q}$$

- u_x
- u_y
- u_z

Control $\mathbf{u}(\tau) = \mathcal{C}(\tau) \bar{\mathbf{u}}$

PARVAL[●] + SPL[SIMTIME[], ●]

▢ Measures

- q_x
- q_y
- q_z

▢ Data Object

▢ parameter

- $\mathbf{f}_{u,x}$
- $\mathbf{f}_{u,y}$
- $\mathbf{f}_{u,z}$

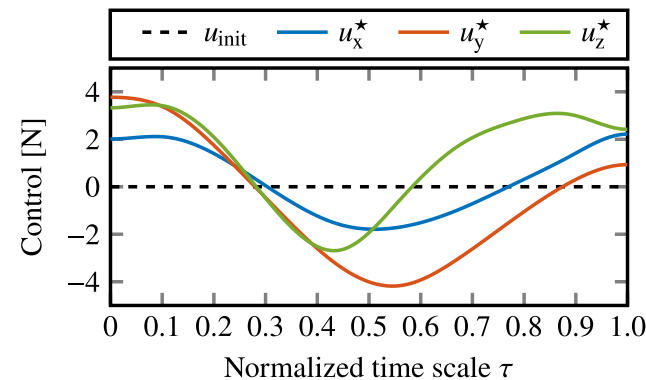
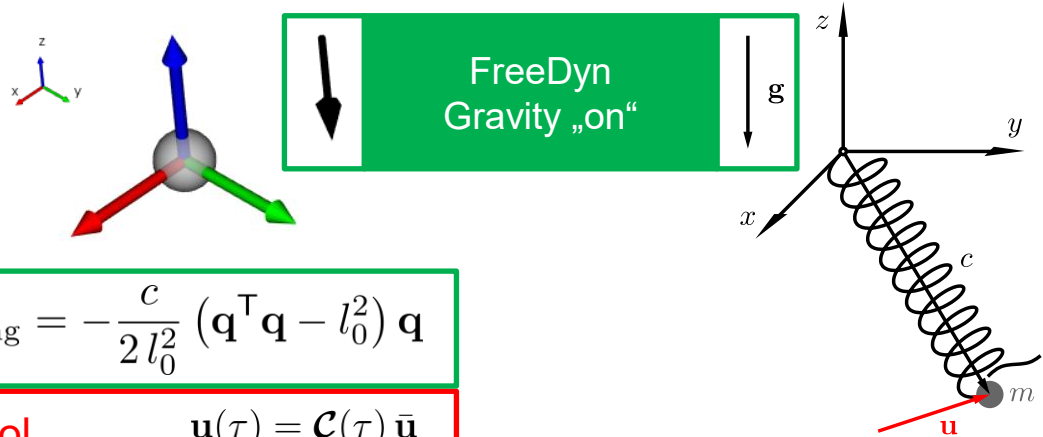
Values: 0

- c
- l_0

▢ spline

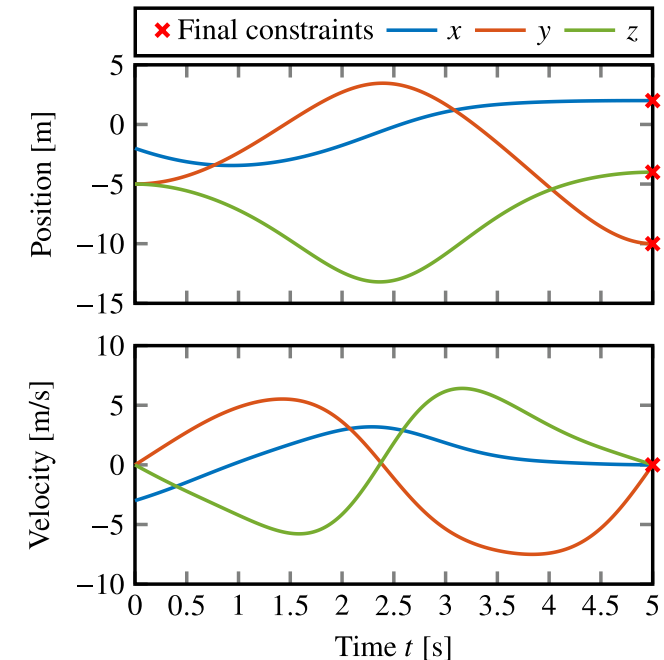
- $\bar{\mathbf{u}}_x$
- $\bar{\mathbf{u}}_y$
- $\bar{\mathbf{u}}_z$

.txt



Objective

$$J = \frac{1}{2} \int_{t_0}^{t_f} \mathbf{u}^T \mathbf{u} dt \quad \phi = \begin{pmatrix} \mathbf{q} - \mathbf{x}_f \\ \dot{\mathbf{q}} \end{pmatrix} \Big|_{t_f}$$



Example – Two Axis SCARA

Rest-To-Rest maneuver with defined final constraints in minimum time

Rigid Bodies

- Body_1
- Body_2
- TCP

Constraints

generic

- Ground <> Body_1
- Body_1 <> Body_2
- Body_2 <> TCP

Forces/Torques

torques generic

- u_1
- u_{2_actio}
- $u_{2_reactio}$

Control

$$\mathbf{u}(\tau) = \mathcal{C}(\tau) \bar{\mathbf{u}}$$

PARVAL[●] + SPL[SIMTIME[],●]

Measures

- φ_1
- φ_2
- ω_1
- ω_2

Data Object

parameter

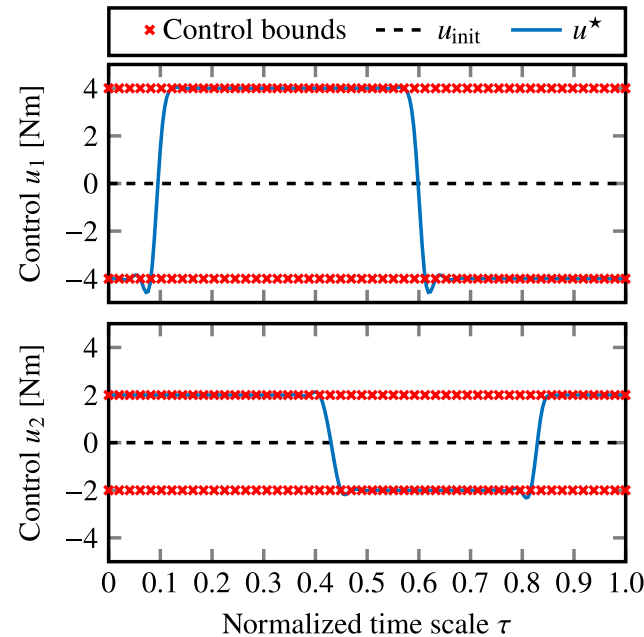
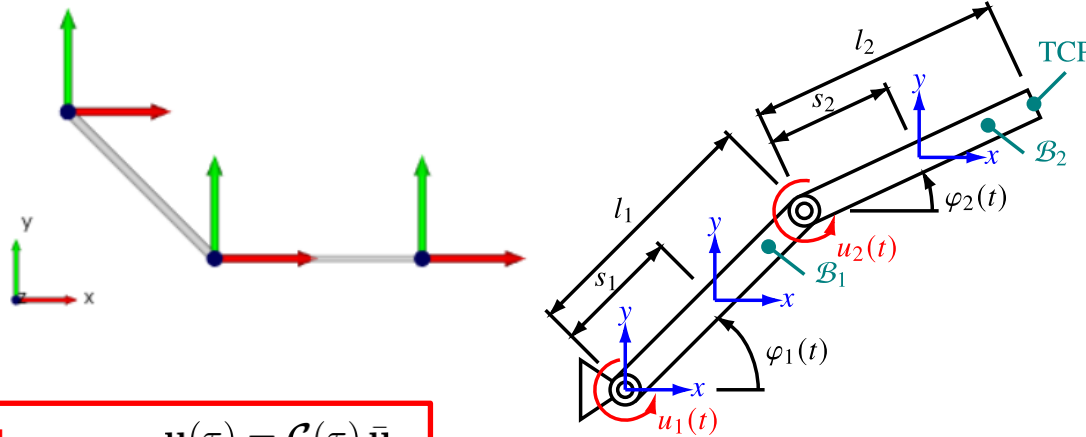
- $\mathbf{f}_{u,1}$
- $\mathbf{f}_{u,2}$

Values: 0

spline

- $\bar{\mathbf{u}}_1$
- $\bar{\mathbf{u}}_2$

.txt



Objective
$J = \int_{t_0}^{t_f} 1 dt \quad \phi = \left(\begin{array}{c} \mathbf{x}_{TCP} - \mathbf{x}_f \\ \dot{\mathbf{x}}_{TCP}(t) \end{array} \right) \bigg _{t_f}$

$$t_f^* = 1.834 s$$

